

# Research Notes on AlphaFold

AlphaFold

## >> Section 1

The 2020 version of the program (AlphaFold 2, 2020) is significantly different from the original version that won CASP 13 in 2018, according to the team at DeepMind. AlphaFold 1 used a number of separately trained modules to produce a guide potential, which was then combined with a physics-based energy potential. AlphaFold 2 replaced this with a system of interconnected sub-networks, forming a single, differentiable, end-to-end model based on pattern recognition. This model was trained in an integrated manner. After the neural network's prediction converges, a final refinement step applies local physical constraints using energy minimization based on the AMBER force field. This step only slightly adjusts the predicted structure. A key part of the 2020 system are two modules, believed to be based on a transformer design, which are used to progressively refine a vector of information for each relationship (or "edge" in graph-theory terminology) between an amino acid residue of the protein and another amino acid residue (these relationships are represented by the array shown in green); and between each amino acid position and each different sequences in the input sequence alignment (these relationships are represented by the array shown in red). Internally these refinement transformations contain layers that have the effect of bringing relevant data together and filtering out irrelevant data (the "attention mechanism") for these relationships, in a context-dependent way, learned from training data. These transformations are iterated, the updated information output by one step becoming the input of the next, with the sharpened residue/residue information feeding into the update of the residue/sequence information, and then the improved residue/sequence information feeding into the update of the residue/residue information. As the iteration progresses, according to one report, the "attention algorithm ... mimics the way a person might assemble a jigsaw puzzle: first connecting pieces in small clumps—in this case clusters of amino acids—and then searching for ways to join the clumps in a larger whole." The output of these iterations then informs the final structure prediction module, which also uses transformers, and is itself then iterated. In an example presented by DeepMind, the structure prediction module achieved a correct topology for the target protein on its first iteration, scored as having a GDT\_TS of 78, but with a large number (90%) of stereochemical violations – i.e. unphysical bond angles or lengths. With subsequent iterations the number of stereochemical violations fell. By the third iteration the GDT\_TS of the prediction was approaching 90, and by the eighth iteration the number of stereochemical violations was approaching zero. The training data was originally restricted to single peptide chains. However, the October

2021 update, named AlphaFold-Multimer, included protein complexes in its training data. DeepMind stated this update succeeded about 70% of the time at accurately predicting protein-protein interactions.

## >> Section 2

The AlphaFold Protein Structure Database (AFDB), a joint project between AlphaFold and EMBL-EBI, was launched on July 22, 2021. At launch, the database contained AlphaFold 1-predicted models for nearly the complete UniProt proteome of humans and 20 model organisms, totaling over 365,000 proteins. The database does not include proteins with fewer than 16 or more than 2700 amino acid residues, but for humans they are available in the whole batch file. AlphaFold's initial goal (as of early 2022) was to expand the database to cover most of the UniRef90 set, which contains over 100 million proteins. As of May 15, 2022, the database contained 992,316 predictions. In July 2021, UniProt-KB and InterPro has been updated to show AlphaFold predictions when available. On July 28, 2022, the team uploaded to the database the structures of around 200 million proteins from 1 million species, covering nearly every known protein on the planet. The number as of 2024 is 214 million, with 26 million being duplicates (exact sequence matches) of another protein in the database. The predicted structures can differ significantly between duplicates. As of 2025, the AFDB uses AlphaFold 2 for its predictions. All structures produced remain monomeric, but multimeric structures produced by other databases are linked on the page through the 3D-Beacons API. Foldseek, which provides fast and accurate structure searches, is also integrated. Information from AlphaMissense (a tool that uses AlphaFold to predict the outcome of missense mutations) is also integrated.

## >> Section 3

Clones and derivatives A number of AlphaFold clones have also been published, mostly with permissive license terms. Clones for AlphaFold3 include ByteDance's Protenix (Apache 2.0 License), AIQuraishi Laboratory's OpenFold-3 (MIT license), and Boltz-1/2 (MIT license). There are also clones for older versions, though they became less relevant with the open-source release of AlphaFold 1 and 2 source codes. Still relevant are models, both open- and closed-source, that include modifications to the AlphaFold architecture. For AlphaFold 2, a notable example is ESMFold from Meta, which replaces the multiple sequence alignment with the latent space of a protein language model. Open-source tools that complement AlphaFold have also been made. One well-cited example is ColabFold, which uses MMseqs2 instead of HHblits to speed up the sequence search,

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allowing the AlphaFold pipelines to run quickly on Google Colab.

## >> Section 4

Published works Andrew W. Senior et al. (December 2019), "Protein structure prediction using multiple deep neural networks in the 13th Critical Assessment of Protein Structure Prediction (CASP13)", *Proteins: Structure, Function, Bioinformatics* 87(12) 1141–1148 doi:10.1002/prot.25834 Andrew W. Senior et al. (15 January 2020), "Improved protein structure prediction using potentials from deep learning", *Nature* 577 706–710 doi:10.1038/s41586-019-1923-7 John Jumper et al. (December 2020), "High Accuracy Protein Structure Prediction Using Deep Learning", in *Fourteenth Critical Assessment of Techniques for Protein Structure Prediction (Abstract Book)*, pp. 22–24 John Jumper et al. (December 2020), "AlphaFold 2". Presentation given at CASP 14. Abramson, J., Adler, J., Dunger, J. et al. (May 2024), "Accurate structure prediction of biomolecular interactions with AlphaFold 3", *Nature* 630, 493–500 (2024)

## >> Section 5

AlphaFold 3 Across several benchmarks, AlphaFold3 has demonstrated, on average, superior performance to conventional search-based docking algorithms in predicting small-molecule–protein binding modes. AlphaFold 3 version can predict structures of protein complexes with a very limited set of selected cofactors and co- and post-translational modifications. Between 50% and 70% of the structures of the human proteome are incomplete without covalently-attached glycans. Studies have shown that although AlphaFold3 can jointly model protein–ligand co-folding, its accuracy drops markedly on test cases with low similarity to its training data—an area of particular importance for drug discovery. Other work has found that AlphaFold is insensitive to adversarial decoys generated by altering the physicochemical properties of binding pockets, suggesting potential reliance on training-set memorization rather than genuine chemical awareness.